

Govt. Post Graduate College for Women
BS Islamiyat S-2.

Course Title: Quantitative Reasoning 1

Course Code: GQR - 101

Submitted by: Group #3

Name

Roll no:

Shehar Bano

148863

Sehar Hameed

148864

Atika Hijazi

148865

Falak Akmal

148866

Malikha Shoukat

148867

Submitted to:

Maam Umme Habiba

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$x = 7$

$y = 9$

Exercise #1

$$x - y = 3$$

$$x + y = 5$$

$$x - y = 3$$

$$x - 3 = y$$

$$x + y = 5$$

$$x + 5 = -y$$

Adding two original equation together-

$$(x - y) + (x + y) = 3 + 5$$

$$2x = 8 \Rightarrow x = \frac{8}{2} = \boxed{x = 4}$$

Σ'

$$4 + y = 5$$

$$y = 5 - 4$$

$$\boxed{y = 1}$$

$$x = 4, y = 1$$

$$2x + 3y = 1$$

$$3x - y = 7$$

$$3y = -2x + 1 \Rightarrow y = \frac{2}{1}$$

$$y = -3x + 7 \Rightarrow x = \frac{3}{7}$$

$$(2x + 3y) + (3x - y) = 1 + 7$$

$$7x + 9y$$

$$\boxed{x = 7}$$

$$\boxed{y = 9}$$

Q3:- $x + 2y = 6$

$2x + 4y = 12$

$x + 2y = 6 \Rightarrow 2y = -x + 6$

$2x + 4y = 12 \Rightarrow 4y = -2x + 12$

$2y = -x + 6$

$4y = x \times 2$

12^6

$$\boxed{4y = \frac{x \times 2}{6}}$$

$= (x + 2y) + (2x + 4y) = 6 + 12$

$= 9x + 18y$

$\boxed{x = 9}$

$\boxed{y = 18}$

Q4:- $3x + 2y = 6$

$6x + 4y = 18$

$3x + 2y = 6 \Rightarrow 2y = -3x + 6$

$6x + 4y = 18 \Rightarrow 4y = -6x + 18$

$4y = x \times 6$

18^3

$= (3x + 2y) + (6x + 4y) = 6 + 18$

$= 15x + 24y$

$\boxed{x = 15}$

$\boxed{y = 24}$

$$\begin{aligned}
 5: - \quad & 2x + y = 5 \\
 & 3x + 2y = 8 \\
 & y = -2x + 5 \\
 & 2y = -3x + 10 \\
 & 2y = x - 3 \\
 & \quad \quad \quad 8
 \end{aligned}$$

$$\begin{aligned}
 & = (2x + y) + (3x + 2y) \quad 5 + 8 \\
 & = 10x + 11y
 \end{aligned}$$

$$x = 10$$

$$y = 11$$

Exercise # 2

Solve the following

$$1) \quad x + y = 14$$

$$x - y = 2$$

Let $y = 14 - x$

put in eq (2)

$$x - 14 + x = 2$$

$$2x - 14 = 2$$

$$x = \frac{16}{2} \quad \boxed{x = 8}$$

put in eq (1)

$$8 + y = 14$$

$$y = 14 - 8$$

$$\boxed{y = 6}$$

$$8 + 6 = 14$$

$$14 = 14$$

$$8 - 6 = 2$$

$$2 = 2$$

Hence proved

$$2) \quad 2x + 3y = 11$$

$$2x - 4y = -24$$

Let y

$$2x = 11 - 3y$$

$$x = \frac{11 - 3y}{2}$$

put in eq ②

$$2x \left[\frac{11 - 3y}{2} \right] - 4y = -24$$

$$11 - 3y - 4y = -24$$

$$-7y = -35$$

$$y = \frac{-35}{-7}$$

$$\boxed{y = 5}$$

Let x

$$x = \frac{11 - 3(5)}{2}$$

$$2$$

$$x = \frac{11 - 15}{2}$$

$$x = \frac{-4}{2}$$

$$x = -2$$

For Check

$$2(-2) + 3(5) = 11$$

$$11 = 11$$

$$2(-2) - 4(5) = -24$$

$$-24 = -24$$

Hence proved

$$3 \quad 3x + 2y = 11$$

$$2x + 3y = 4$$

$$y = \frac{11 - 3x}{2} \quad \text{--- (iii)}$$

Substituting the value of y from (iii) in (ii) we get

$$2x + 3 \left[\frac{11 - 3x}{2} \right] = 4$$

$$4x + 33 - 9x = 8$$

$$-5x = -25$$

$$x = 5$$

putting $x = 5$ in (iii) we get

$$y = \frac{11 - 3 \times 5}{2} = \frac{11 - 15}{2}$$

$$y = \frac{-4}{2}$$

$$y = -2$$

6

$$\text{So, } x=5 \text{ \& } y=-2$$

Q4:-

$$7x - 2y = 1$$

$$3x + 4y = 15$$

$$\text{eqn (1) } 7x - 2y = 1$$

$$7x = 1 + 2y$$

$$x = \frac{1+2y}{7}$$

$$\text{eqn (2) } 3x + 4y = 15$$

$$3\left(\frac{1+2y}{7}\right) + 4y = 15$$

$$\frac{3+6y}{7} + 4y = 15$$

$$\frac{3+6y+28y}{7} = 15$$

$$3+34y = 15 \times 7$$

$$34y = 105 - 3$$

$$y = \frac{102}{34}$$

$$y = 3$$

$$y = 3$$

$$x = \frac{1+2 \times 3}{7} = \frac{7}{7} = 1$$

$$7$$

$$7$$

$$\text{So } x=1 \text{ and } y=3$$

Exerc

Solve the
by eliminate

$$3x + 4y$$

$$3x - y =$$

Subtract

$$3x$$

$$- 3x$$

$$y = -$$

$$(x, y)$$

$$-1- x$$

$$2x$$

Multi

1:- $3x + 4y = -6$ eq (i)

$$3x - y = 9 \quad \text{eq (ii)}$$

Subtract eq (i) - (ii)

$$\begin{array}{r} 3x + 4y = -6 \\ \ominus 3x - y = 9 \\ \hline 5y = -15 \end{array}$$

$$y = \frac{-15}{5}$$

$$\boxed{y = -3}$$

$y = -3$ put in eq (ii)

$$3x - y = 9$$

$$3x + 3 = 9$$

$$3x = 9 - 3$$

$$3x = 6$$

$$x = \frac{6}{3} \quad \boxed{x = 2}$$

$$(x, y) = (2, -3)$$

2:- $x + 2y = 5$

$$2x + 3y = 8$$

Multiply 2 by eq (i)

$$2x + 4y = 10$$

$$\begin{array}{r} 2x + 4y = 10 \\ \ominus 2x + 3y = 8 \\ \hline y = 2 \end{array}$$

$$\boxed{y = 2}$$

Date: _____

Day: _____

$y = 2$ put in eq (i)

(2)

Mult

Sub

$$x + 2y = 5$$

$$x + 2(2) = 5$$

$$x + 4 = 5$$

$$x = 5 - 4$$

$$\boxed{x = 1}$$

$$(x, y) = (1, 2)$$

3:-

$$x - 2y = 7$$

$$3x + y = 35$$

Multiply 3 by eq (i)

$$3x = 6y = 21 \quad \text{--- (ii)}$$

Sub eq (ii) - (i)

$$3x + y = 35$$

$$\ominus \quad 3x \oplus 6y \ominus 21$$

$$7y = 14$$

$$y = \frac{14}{7} \quad \boxed{y = 2}$$

$y = 2$ put in eq (i)

$$x - 2(2) = 7$$

$$x - 4 = 7$$

$$x = 7 + 4$$

$$x = 11$$

$$(x, y) = (11, 2)$$

4:-

$$3x + 4y = 15$$

$$7x - 2y = 1$$

Date: _____
 Multiply 2 by eq (i) (ii)
 $14x - 4y = 2$ (iii)

Add eq (i) + (ii)

$$3x + 4y = 15$$

$$14 - 4y = 2$$

$$17x = 17$$

$$x = 1$$

$x = 1$ put eq (ii)

$$7(1) - 2y = 1$$

$$7 - 2y = 1$$

$$-2y = 1 - 7$$

$$+2y = -6$$

$$y = \frac{-6}{2}$$

$$y = -3$$

$$(x, y) = (1, -3)$$

Q5:-

$$2x + 3y = 4$$

$$3x + 2y = 11$$

Multiply 3 by eq (i)

$$6x + 9y = 12$$
 (iii)

Multiply 2 by eq (ii)

$$6x + 4y = 22$$

$$6x + 9y = 12$$

$$-5y = 10$$

$$y = \frac{-10}{5}$$

$$\boxed{y = -2}$$

-2 put eq (i)

$$2x + 3(-2) = 4$$

$$2x - 6 = 4$$

$$2x = 4 + 6$$

$$2x = 10$$

$$x = \frac{10}{2}$$

$$\boxed{x = 5}$$

$$(x, y) (5, -2)$$

$$3x - 5y = 23$$

$$2x - 4y = 16$$

Multiply 2 by eq (i)

$$6x - 10y = 46 \quad \text{--- (iii)}$$

Multiply 3 by eq (ii)

$$6x - 12y = 48 \quad \text{--- (iv)}$$

Sub eq (iv - iii)

$$6x - 12y = 48$$

$$6x - 10y = 46$$

$$\begin{array}{r} \ominus \quad 6x - 12y = 48 \\ \oplus \quad 6x - 10y = 46 \\ \hline -2y = 2 \end{array}$$

$$\boxed{y = 1}$$

y=1 put eq (ii)

$$\begin{aligned}
 2x - 4(-1) &= 16 \\
 2x + 4 &= 16 \\
 2x &= 16 - 4 \\
 2x &= 12 \\
 x &= \frac{12}{2}
 \end{aligned}$$

$$x = 6$$

(x, y) (6, 1)

Exercise # 4

Q1:- Rahim's father is three times as old as Rahim. If sum of their ages is 56 years find their ages?

Let the age of Rahim x years and his father's age is $3x$ years.

$$x + 3x = 56$$

$$4x = 56$$

$$x = \frac{56}{4}$$

$$x = 14$$

$$x = 14$$

So Rahim's age is 14 years old.

and his father is $3 \times 14 = 42$

Rita has 18 meter cloth. She cut it into two pieces in such a way

2 = 16
16
4

#4

times as old
their ages is

years and
ars.

d.

7

cut

wa

Date:

Day:

one piece is 4m longer than other.
What is the length of shorter pieces?

Shorter meter = x

Longer meter = $x + 4$

$$x + (x + 4) = 18$$

$$2x + 4 = 18$$

$$2x = 18 - 4$$

$$2x = 14$$

$$x = \frac{14}{2}$$

$$x = 7$$

$$x = 7$$

Length of the shorter piece is 7 meter.

A total of Rs 50,000 is to be distributed among 200 persons as prizes.

A prize is either Rs 500 or Rs 100.

Find the number of each type of prizes?

500 prizes be x

100 prizes be $(200 - x)$

$$500x + 100(200 - x) = 50,000$$

$$400x + 20,000 = 50,000$$

$$400x = 50,000 - 20,000$$

$$400x = 30,000$$

$$x = \frac{30,000}{400}$$

$$x = 75$$

$$x = 75$$

Day: _____

Rs prizes is
of Rs 100 prizes

Date: _____

Day: _____

So the number of Rs 50 notes is
16 and the number of Rs 100 notes is -

$$x+1 = 16+1 = 17$$

Final answer

Rs 50 note = 16

Rs 100 notes = 17.

100 in notes
and 50 - If
notes is one

Notes find
each denomination?

$$x+1$$

purse is
write the eq.

100